

AC-04: A certified upper-bound refinement

Iterated slice extraction for the small Coppersmith–Winograd tensor

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Scope and status. This document does *not* prove or disprove $\tilde{R}(\text{cw}_2) = 3$. It derives the strict upper bound $\tilde{R}(\text{cw}_2) < 3.923038$ from the speedup framework of Alman and Li, with an analytic parameter argument and exact rational certificates. The underlying speedup method is existing work. No claim of research priority, independent peer review, or proof-assistant verification is made.

Abstract

For the tensor in repository problem AC-04, a rank-four normal form supplies a full, isolated slice degeneration starting from the third tensor power. Repeated squaring and re-extraction lead to a closed recurrence for dimensions and spectral constraints. An analytic monotonicity lemma reduces the entire relevant spectral parameter interval to the endpoint $2/3$. Three re-extractions, certified using integer cube roots and rational interval arithmetic, give $3 \leq \tilde{R}(T) < 3.923038$; the scalar endpoint is approximately 3.923037967885872253 . The same recurrence, even with arbitrarily many iterations, allows a formal scalar value of 3.9 . This last observation is an obstruction *only to deductions from these scalar constraints*, not a lower bound on $\tilde{R}(T)$. The original equality question remains unproved in this work.

1 The question and the result actually established

The tensor is

$$T = \sum_{i=1}^2 (e_0 \otimes e_i \otimes e_i + e_i \otimes e_0 \otimes e_i + e_i \otimes e_i \otimes e_0) \in \mathbb{C}^3 \otimes \mathbb{C}^3 \otimes \mathbb{C}^3. \quad (1)$$

All tensor powers in this document are grouped, three-mode Kronecker powers: the first factors are grouped together, and likewise the second and third. The question is whether

$$\tilde{R}(T) := \lim_{k \rightarrow \infty} R(T^{\otimes k})^{1/k} = 3. \quad (2)$$

This is the exact target in AC-04 [1], not the assertion that some fixed power has rank or border rank 3^k .

Theorem 1.1 (Upper-bound refinement). *For the tensor (1),*

$$3 \leq \tilde{R}(T) < 3.923038.$$

More precisely, $\tilde{R}(T) \leq \beta$, where β is the unique nonnegative solution of the scalar equation in Section 7, and

$$3.92303796788587225299 < \beta < 3.92303796788587225301.$$

The source benchmark is the bound $\tilde{R}(T) < 3.931$ of Alman and Li [2]. Their framework explicitly permits further iteration. The present calculation changes the initial tensor power to three and makes the iteration and endpoint certification explicit; it is not a new speedup theorem. Their Table 1, using initial power four for the quoted second-method bound, reports 3.930872. This comparison is to the displayed benchmark, not an assertion that all unpublished or subsequent optimizations have been ruled out.

The equality in (2) is known to imply $\omega = 2$ [2, Theorem 1.2]. We do not use that implication to dismiss the problem, and we do not assert its converse. The stronger equality is simply not a conclusion of the proof below.

2 An exact rank-four normal form

Let

$$P = \sum_{\sigma \in S_3} a_{\sigma(1)} \otimes a_{\sigma(2)} \otimes a_{\sigma(3)},$$

where a_1, a_2, a_3 is the standard basis of $\mathbb{C}^3$. Set $w_1 = e_0$, $w_2 = e_1 + ie_2$, and $w_3 = e_1 - ie_2$. Then

$$w_2 \otimes w_3 + w_3 \otimes w_2 = 2(e_1 \otimes e_1 + e_2 \otimes e_2),$$

so, for the invertible map $La_i = w_i$,

$$T = \frac{1}{2}L^{\otimes 3}P. \quad (3)$$

Nonzero scalar factors and invertible changes of basis preserve rank, border rank, and asymptotic rank. It is therefore enough to work with P .

Define

$$v_0 = (1, 1, 1)^T, \quad v_1 = (1, -1, -1)^T, \quad v_2 = (-1, 1, -1)^T, \quad v_3 = (-1, -1, 1)^T.$$

A coefficient check gives the exact decomposition

$$P = \frac{1}{4} \sum_{\ell=0}^3 v_\ell \otimes v_\ell \otimes v_\ell. \quad (4)$$

Indeed, the coefficient is one when all three indices differ, and zero otherwise. In particular, $R(T) \leq 4$.

The three slices of a flattening of P are linearly independent: each has its nonzero entries on a different pair of off-diagonal positions. Thus every flattening of P has rank three. Flattening commutes with the grouped tensor product, and matrix ranks multiply. Consequently,

$$R(T^{\otimes k}) \geq 3^k \quad (k \geq 1).$$

Submultiplicativity of tensor rank makes $\log R(T^{\otimes k})$ subadditive; its normalized limit is its infimum. This proves both existence of the asymptotic-rank limit and the lower bound in Theorem 1.1.

For extraction, let $V = [v_0 \ v_1 \ v_2 \ v_3]$ and $D = \text{diag}(1, 1, -1, -1)$. Direct multiplication gives

$$VDV^T = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 4 \\ 0 & 4 & 0 \end{pmatrix} = XY^T, \quad X = \begin{pmatrix} 0 & 0 \\ 4 & 0 \\ 0 & 4 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (5)$$

All four diagonal coefficients in D are nonzero, while the contracted image has rank two. This is the finite algebra needed below.

3 The extraction mechanism and its hypotheses

We spell out the relevant mechanism so the numerical recurrence is not treated as a free-standing conjecture. It is the one-slice specialization of the free-lunch construction in [2, Section 5].

Write

$$D_r = \sum_{i=1}^r a_i \otimes b_i \otimes c_i, \quad L_s = \sum_{i=1}^s u_i \otimes v_i \otimes w.$$

Thus D_r is diagonal and L_s has dimensions $(s, s, 1)$; in matrix multiplication notation it is $\langle 1, s, 1 \rangle$. Put $L_0 = 0$. The notation $K \leq S$ means that K is a degeneration of S , with maps allowed to have rational-function coefficients in a degeneration parameter.

3.1 A kernel-compression lemma

Lemma 3.1. *Suppose maps A, B, C send a tensor S to K , and the first two target dimensions are d_A, d_B . Let f be a functional on the third source space such that*

$$(A \otimes B \otimes f)S = 0.$$

If the source contraction $H = (\text{id} \otimes \text{id} \otimes f)S$ has matrix rank M , then S degenerates to $K \oplus L_t$ for every positive integer $t \leq M - d_A - d_B$. The new slice is isolated in its third mode. Its source fullness index is M . The statement also applies over a rational-function field and can be composed with a degeneration to K .

Proof. Regard rows of A, B as functionals on the respective source spaces. Let

$$E = \{u : uHB^\top = 0\}, \quad F = \{v : AHv^\top = 0\}.$$

Their codimensions are at most d_B, d_A . Restricting a matrix to a subspace of codimension c on either side decreases its rank by at most c . Hence the restriction of H to $E \times F$ has rank at least $M - d_A - d_B$. Choose row maps A', B' from these subspaces and then restrict and normalize a rank- t submatrix so that $(A' \otimes B' \otimes f)S = L_t$.

Augment the three maps to $(A, A'), (B, B')$, and (C, f) . There are eight blocks indexed by 1, 2 in the three modes. The 111 block is K and the 222 block is L_t . By construction, blocks 112, 122, 212 vanish identically. Assign weights $(1, \lambda)$ to the two first-mode blocks, the same weights to the second-mode blocks, and weights $(\lambda^2, 1)$ to the third-mode blocks. Divide the resulting tensor by λ^2 . The 111 and 222 blocks survive. Blocks 121 and 211 have an extra factor λ , and block 221 an extra factor λ^2 ; they disappear. This proves the degeneration.

Every block ending in the new third-mode coordinate, other than 222, is identically zero even before taking the limit. This is isolation. The functional selecting that coordinate is a nonzero scalar multiple of f , so its contraction on the *source* still has rank M ; this is fullness index M .

For rational-function input maps, perform these linear-algebra operations over that field. Normalize the extracted rank- t matrix before taking the limit, so that rank is not lost in its leading coefficient. The finitely many entries of all maps have finite pole orders. Replacing the new parameter by a sufficiently high power of the original one makes every unwanted block tend to zero. The desired K block retains its original limit, and the normalized slice stays constant. Isolation remains exact, and source contraction rank is unchanged over the rational-function field. $\square$

3.2 Why full and isolated are essential

For a degeneration $K \oplus L_t \trianglelefteq S$, with maps $A_\lambda, B_\lambda, C_\lambda$, call the distinguished L_t *isolated* if the third-mode coordinate selecting it has identically zero coefficients outside its own first/second-mode block before the limit. Call it *full* when the associated contraction of S has rank equal to each of the first two source dimensions. These are properties of the maps defining the degeneration, not merely of the limiting direct sum.

Tensor products preserve both properties: the relevant source contraction is a Kronecker product, so ranks multiply; and an off-block contribution to the product's distinguished third coordinate contains an identically zero factor. This is the mechanism also formalized in [2, Propositions 5.1–5.6]. Deleting a full, isolated slice leaves its source functional satisfying the zero-contraction hypothesis of Lemma 3.1. Without isolation, an $O(\lambda)$ cross-block term would not suffice for this argument.

4 The base degeneration: dimensions 27, 18, and 72

Use the rank-four decomposition (4) three times. Thus $P^{\otimes 3}$ is a restriction of D_{64} with first two maps $A = V^{\otimes 3}$ and third map $4^{-3}V^{\otimes 3}$. The source third-mode functional with coefficient matrix $D^{\otimes 3}$ has 64 nonzero entries, and its image contraction is

$$AD^{\otimes 3}A^\top = (VDV^\top)^{\otimes 3} = X^{\otimes 3}(Y^{\otimes 3})^\top,$$

of rank $2^3 = 8$.

Adjoin L_8 to the source. Map its first two modes by $X^{\otimes 3}$ and $Y^{\otimes 3}$, and its third mode by zero for the map onto $P^{\otimes 3}$. Extend the chosen third functional with coefficient -1 on the L_8 slice. The contracted image cancels exactly. On the source, the contraction matrix is

$$H = D^{\otimes 3} \oplus (-I_8),$$

which is nonsingular of size 72. Lemma 3.1, with $d_A = d_B = 27$, yields a full and isolated degeneration

$$U_0 := P^{\otimes 3} \oplus L_{18} \trianglelefteq S := D_{64} \oplus L_8, \quad 18 = 72 - 2 \cdot 27. \quad (6)$$

It holds in all three slice orientations, because P is symmetric in its modes.

The file `code/algebra_check.py` checks the complete finite linear algebra here using exact SymPy matrices. The two annihilator spaces have dimension 45. Their contracted 45×45 matrix has rank 18; the certificate supplies an 18×18 minor with determinant -1 . Thus the guaranteed slice size is realized by this explicit base construction, not inferred from a floating-point numerical rank.

5 Repeated squaring and re-extraction

Proposition 5.1. *There are degenerations*

$$U_j = K_j \oplus L_{t_j} \leq S^{\otimes 2^j} \quad (j \geq 0),$$

whose distinguished slices are full and isolated. The first two dimensions of K_j are d_j , and those of the source are M_j , where

$$\begin{aligned} (d_0, t_0, M_0) &= (27, 18, 72), \\ d_{j+1} &= d_j^2 + 2d_j t_j, \\ t_{j+1} &= t_j^2 + 2d_j^2, \\ M_{j+1} &= M_j^2 = t_{j+1} + 2d_{j+1}. \end{aligned} \tag{7}$$

The third dimension of K_j need not equal d_j .

Proof. The base case is (6). Square the degeneration at step j and separate its distinguished summand $L_{t_j^2}$. The remaining tensor is

$$K_{j+1} = K_j^{\otimes 2} \oplus (K_j \otimes L_{t_j}) \oplus (L_{t_j} \otimes K_j),$$

whose first two dimensions are $d_j^2 + 2d_j t_j$. The removed slice has source fullness M_j^2 and is isolated. Its source functional therefore meets the cancellation condition after its image variables are removed. Re-extract with Lemma 3.1 to obtain

$$\begin{aligned} t_{j+1} &= M_j^2 - 2(d_j^2 + 2d_j t_j) \\ &= (t_j + 2d_j)^2 - 2d_j^2 - 4d_j t_j = t_j^2 + 2d_j^2. \end{aligned}$$

The new slice is again full and isolated. This proves the induction, including $M_{j+1} = t_{j+1} + 2d_{j+1}$. $\square$

This is the repeated form of the mechanism in [2, Theorem 6.2]. There is a useful exact closed form:

$$d_j = \frac{72^{2^j} - (-9)^{2^j}}{3}, \quad t_j = \frac{72^{2^j} + 2(-9)^{2^j}}{3}, \quad M_j = 72^{2^j}. \tag{8}$$

Indeed, both $2d_j + t_j$ and $t_j - d_j$ are squared by the recurrence, with initial values 72 and -9 . The first values are

j	d_j	t_j	M_j
0	27	18	72
1	1,701	1,782	5,184
2	8,955,765	8,962,326	26,873,856
3	240,734,697,754,005	240,734,740,800,726	722,204,136,308,736

The large dimensions are tracked symbolically; no tensor of these dimensions is explicitly allocated in the computation.

6 Turning the degenerations into scalar bounds

We use Strassen's asymptotic-spectrum duality in its tensor form, as stated and applied in [2, Section 4]. Precisely, for complex tensors there is a family of spectral points ϕ such that

$$\tilde{R}(P) = \max_{\phi} \phi(P).$$

A spectral point is additive for direct sums, multiplicative for grouped tensor products, monotone under degeneration, and normalized by $\phi(D_r) = r$. For the three orientations of L_s , its values are s^{θ_1} , s^{θ_2} , s^{θ_3} , with

$$0 \leq \theta_i \leq 1, \quad \theta_1 + \theta_2 + \theta_3 \geq 2. \quad (9)$$

These are standard external theorems used here, not assertions that the attached code proves. In particular, at least one orientation has $\theta \in [2/3, 1]$.

Fix a spectral point and choose that orientation. Write

$$z = \phi(P), \quad x = z^3, \quad h(\theta) = 64 + 8^\theta.$$

The exact rank-four upper bound gives $0 \leq z \leq 4$, hence $0 \leq x \leq 64$. Define

$$\begin{aligned} p_0(x, \theta) &= x + 18^\theta, \\ p_{j+1}(x, \theta) &= p_j(x, \theta)^2 - t_j^{2\theta} + t_{j+1}^\theta. \end{aligned} \quad (10)$$

The subtraction removes precisely the square of the old distinguished slice, which is replaced by the new one. Additivity and multiplicativity give $\phi(U_j) = p_j(x, \theta)$. Consequently every spectral point obeys

$$p_j(x, \theta) \leq h(\theta)^{2^j} \quad (j \geq 0). \quad (11)$$

Although the displayed recursion subtracts a constant, p_j is the value of an actual direct sum: as a polynomial in x , all its coefficients are nonnegative, its constant term is t_j^θ , and it is strictly increasing for $x \geq 0$.

6.1 An analytic endpoint lemma

Numerically searching for the worst θ would not certify an interval of parameters. The next lemma eliminates that search.

Lemma 6.1. *For fixed $x \in [0, 64]$ and each $j \geq 0$, the function*

$$q_j(x, \theta) = \frac{p_j(x, \theta)}{h(\theta)^{2^j}}$$

is strictly increasing for $\theta > 0$.

Proof. For $j = 0$, the numerator of $\partial q_0 / \partial \theta$, after multiplying by the positive denominator h^2 , is

$$64 \cdot 18^\theta \log 18 - x \cdot 8^\theta \log 8 + 18^\theta 8^\theta \log(18/8).$$

It is strictly positive: $x \leq 64$, $18 > 8$, and $\theta > 0$.

Normalize the recursion to obtain

$$q_{j+1} = q_j^2 + E_j(\theta), \quad E_j(\theta) = \frac{t_{j+1}^\theta - t_j^{2\theta}}{h(\theta)^{2^{j+1}}}. \quad (12)$$

We have $t_{j+1} > t_j^2$ and, inductively from $t_0 = 18 > 8$, $t_j > 8^{2^j}$. If $A > B > 0$ and $\theta > 0$, then

$$\frac{d}{d\theta} \log(A^\theta - B^\theta) = \log B + \frac{A^\theta \log(A/B)}{A^\theta - B^\theta} > \log B.$$

On the other hand, $h'/h < \log 8$. Apply these inequalities with $A = t_{j+1}$ and $B = t_j^2$. They imply

$$\frac{E'_j(\theta)}{E_j(\theta)} > 2 \log t_j - 2^{j+1} \log 8 > 0.$$

Thus E_j is positive and strictly increasing. Since q_j is positive, (12) proves the claim by induction. $\square$

For the chosen orientation $\theta \geq 2/3$, equations (11) and Lemma 6.1 imply

$$p_j(z^3, 2/3) \leq 68^{2^j}. \quad (13)$$

This is an *upper* bound: monotonicity in θ makes the endpoint constraint weaker, so every actual spectral point must satisfy it. No permutation-invariance of ϕ was assumed; the symmetry of P supplies three separate oriented degenerations.

7 Exact certificate for the strict bound

Define β by

$$p_3(\beta^3, 2/3) = 68^8. \quad (14)$$

The left side is strictly increasing in $\beta \geq 0$. The certified bracket below proves existence and uniqueness of the nonnegative root. Every spectral value z satisfies (13), so $z \leq \beta$. Duality gives $\tilde{R}(P) \leq \beta$.

Only cube roots of positive integers are needed for evaluation, because $t^{2/3} = \sqrt[3]{t^2}$. For any nonnegative integer N , let

$$a = \left\lfloor \sqrt[3]{N 10^{120}} \right\rfloor.$$

Integer comparisons determine a exactly. Then

$$\frac{a}{10^{40}} \leq \sqrt[3]{N} \leq \frac{a+1}{10^{40}},$$

with equality handled exactly for perfect cubes. Interval addition, subtraction, squaring, and division by positive rational numbers preserve rigorous enclosures.

The standard-library checker `code/certify.py` propagates these intervals through (10). At the rational number $\alpha = 3.923038$, it proves

$$\frac{p_3(\alpha^3, 2/3)}{68^8} > 1.$$

For a readable indication of the margin, the enclosed value is approximately

$$1.0000001724140039588978237928635588.$$

The comparison with one is made using exact rational endpoints, not this rounded decimal. It follows that $\beta < 3.923038$.

The same checker independently proves the two strict inequalities

$$\begin{aligned} p_3((3.92303796788587225299)^3, 2/3) &< 68^8, \\ p_3((3.92303796788587225301)^3, 2/3) &> 68^8. \end{aligned}$$

Both inputs here are terminating rationals. These checks give the bracket in Theorem 1.1. Combining it with the flattening lower bound completes the proof of that theorem.

7.1 Numerical progression and its proper interpretation

For context, the scalar roots for successive values of j are shown below. Rows not used in the exact certificate are exploratory high-precision evaluations, not additional computer-assisted theorems.

Re-extractions j	Initial power $n = 3$: endpoint root
0	3.939328446458942411...
1	3.923413930957849239...
2	3.923038786084391559...
3	3.923037967885872253... (exact bracket certified)
4	3.923037967878664608...
5	3.923037967878664608...

For comparison, starting with $n = 4$ gives endpoint values approximately 3.9334837730 at $j = 0$ and 3.9308711169 at $j = 1$, consistent with the upward-rounded benchmark in [2, Table 1]. The initial power that is best before re-extraction need not remain best after it. No global optimality over initial powers, mixing patterns, or other degeneration methods is claimed.

The step $j = 3$ involves a 24th power of P within a much larger direct sum. This does *not* provide an explicit assertion $R(P^{\otimes 24}) \leq 3.923038^{24}$. The argument passes from direct-sum constraints to asymptotic rank through spectral duality. Confusing these two claims would invalidate the interpretation of the result.

8 A rigorous limitation of this particular recurrence

The rapid apparent numerical stabilization has an analytic counterpart. We give a deliberately modest threshold, 3.9, for which a simple uniform argument works.

Proposition 8.1 (Method-local scalar obstruction). *The constraints (11) for all $j \geq 0$, generated by the $n = 3$ binary re-extraction family in all three orientations and supplemented by (9) and $0 \leq z \leq 4$, admit the formal assignment*

$$z = 39/10, \quad \theta_1 = \theta_2 = \theta_3 = 2/3.$$

Therefore these constraints alone cannot imply $z < 3.9$, and in particular cannot establish the target $\tilde{R}(P) = 3$.

Proof. Put $x = (39/10)^3$, $h = 68$, and $c = 39/40$. Write $q_j = q_j(x, 2/3)$. The exact checker verifies

$$q_0 = \frac{(39/10)^3 + 18^{2/3}}{68} < c, \quad E_0 = \frac{1782^{2/3} - 18^{4/3}}{68^2} < c - c^2 = \frac{39}{1600}.$$

For reference, these values are approximately 0.9733424332 and 0.02158547644. The exact comparisons, not the decimal displays, are used.

For $j \geq 1$, formula (8) gives $t_{j+1} \leq 72^{2^{j+1}}$, hence

$$\begin{aligned} 0 < E_j(2/3) &< \frac{t_{j+1}^{2/3}}{68^{2^{j+1}}} \leq \left(\frac{72^{2/3}}{68} \right)^{2^{j+1}} \\ &< \left(\frac{9}{34} \right)^4 < \frac{39}{1600}. \end{aligned}$$

Here $72^{2/3} < 18$ follows from the integer inequality $72^2 < 18^3$. The last comparison is rational and exact. If $q_j < c$, the recurrence gives $q_{j+1} = q_j^2 + E_j < c^2 + (c - c^2) = c$. Thus $q_j < c < 1$ for every j . All orientations have the same chosen exponent, so all displayed constraints are satisfied, as are $\theta_1 + \theta_2 + \theta_3 = 2$ and $z \leq 4$. $\square$

This is not a tensor lower bound. We have not constructed a spectral point with $\phi(P) = 3.9$. A feasible assignment for a subset of the necessary spectral inequalities need not extend to a genuine spectral point on all tensors. Accordingly, Proposition 8.1 says nothing of the form $\tilde{R}(P) \geq 3.9$. Its use is to identify a precise missing ingredient: more iterations of these same scalar inequalities cannot close the gap to three.

9 What remains unproved, and two structural reformulations

A complete positive solution still needs to establish the quantified statement

$$\text{for every } \varepsilon > 0 \text{ there exists } k \geq 1 \text{ such that } R(T^{\otimes k}) < (3 + \varepsilon)^k. \quad (15)$$

By submultiplicativity this is equivalent to (2). Equivalently, one needs the upper growth estimate $R(T^{\otimes k}) = 3^{k+o(k)}$. Replacing rank by border rank leads to the same asymptotic target, using the standard rank/border-rank asymptotic relation discussed in [2]. No such estimate is supplied here.

Two elementary descriptions locate the issue without asserting that it is solved. First, identify the three indices with the three nonzero elements of $G = (\mathbb{Z}/2\mathbb{Z})^2$. Then the support of P is exactly

$$\{(a, b, c) \in (G \setminus \{0\})^3 : a + b + c = 0\}.$$

Thus $P^{\otimes k}$ is a restricted group-convolution tensor. The ambient group has 4^k elements while each restricted mode has 3^k elements. An algorithm with $3^{k+o(k)}$ bilinear multiplications for this restricted convolution would prove the desired upper bound. The ambient Fourier decomposition merely recovers the four-term construction; it does not provide that improvement. This support identification follows by checking the three possible nonzero values in one coordinate.

Second, index a cyclic group by 0, 1, 2 and write

$$C_3 = \sum_{a+b+c \equiv 0 \pmod{3}} e_a \otimes e_b \otimes e_c.$$

Its nine terms consist of the six terms of P (with indices relabeled) and three diagonal terms. Hence

$$P = C_3 - D_3.$$

A Fourier decomposition gives $R(C_3) = 3$. It is not valid to infer $\tilde{R}(C_3 - D_3) \leq 3$ from this identity: subtraction is not a tensor restriction, and asymptotic rank has no subtraction rule.

There is even a small obstruction to a tempting diagonal degeneration of this identity. Suppose local monomial weights a_i, b_j, c_k preserve all six permutation terms at a common total weight m . Swapping indices in those terms forces $a_i - b_i$ to be constant, and likewise $a_i - c_i$ to be constant. The weights of the three unwanted diagonal terms then have sum $3m$. They cannot all be strictly greater than m , so this local weighting alone cannot kill all three diagonal terms while preserving all six desired ones. This observation excludes only this elementary diagonal-weighting attempt; it does not exclude general degenerations, restrictions of larger powers, or a positive solution to AC-04.

Finally, the known finite result $\underline{R}(T^{\otimes 2}) = 16$ [3] cannot be promoted to $\tilde{R}(T) = 4$. The asymptotic rank is an infimum over all powers. Neither this finite lower bound nor the bound proved in this report resolves which point in the surviving interval is the asymptotic rank.

10 Audit, reproducibility, and delivery scope

The theorem proof uses an existing asymptotic-spectrum theorem and the slice-extraction mechanism, with their exact roles stated in Sections 3–6. The new calculations are the initial parameter choice, explicit repeated recurrence, all-parameter endpoint proof, exact bound certificate, and the limited scalar obstruction. Priority for these refinements has not been established by the literature search.

The scalar certificate uses only the Python standard library. The separate finite-matrix audit requires SymPy; it was run with version 1.14.0. The following commands reproduce the supplied checks from the package root:

```
python3 code/certify.py --output results/exact_certificate.json
python3 code/algebra_check.py --output results/algebra_certificate.json
(cd code && python3 -m unittest -v test_certify.py)
```

The delivered run passes eight regression tests and both exact certificate programs. The main numeric test uses no floating-point comparison. The high-precision exploratory table is generated separately and is not used as a proof of a strict inequality.

No proof assistant was run. No independent referee has audited the full argument. Passing the algebra and interval checks is not equivalent to either formal verification or outside peer review. The local status of this package is **partial result**; the repository itself was not edited, and this package must not be reported as a solution or disproof of AC-04. The repository’s resolution guidance likewise distinguishes supporting results from a complete resolution of the original target [4].

References

- [1] OpenProblemsInNLA, *AC-04: Asymptotic rank of the small Coppersmith–Winograd tensor*, canonical problem entry, consulted for this request.
<https://github.com/ajt60gaibb/OpenProblemsInNLA/tree/main/arithmetic-and-complexity/AC-04>
- [2] J. Alman and B. Li, *Asymptotic Rank Speedup Theorems, Revisited*. 41st Computational Complexity Conference (CCC 2026), LIPIcs 383, 36:1–36:42, published 23 July 2026. DOI: [10.4230/LIPIcs.CCC.2026.36](https://doi.org/10.4230/LIPIcs.CCC.2026.36). Detailed theorem locators in this report refer to the arXiv v1 exposition: Theorem 4.1 and Propositions 4.1–4.2 (spectrum), Sections 5.2–5.3 (extraction, fullness and isolation), Theorems 6.1–6.2 (speedups), and Section 7.1/Table 1 (CW bounds).
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- [3] A. Conner, H. Huang, and J. M. Landsberg, *Bad and good news for Strassen’s laser method: Border rank of the 3×3 permanent and strict submultiplicativity*, arXiv:2009.11391 (2020), Theorem 1.1.
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- [4] OpenProblemsInNLA, *Contributing*, especially “Reporting a resolution” and the distinction between supporting lemmas and complete target verification.
<https://github.com/ajt60gaibb/OpenProblemsInNLA/blob/main/CONTRIBUTING.md>